SING's Cross-Model Leakage Ranking Is Indistinguishable from a Null-Energy Ranking

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Abstract

SING (Yadid et al., 2026) ranks 13 ImageNet classifiers by how much class semantics "leak" into the null space of their linear heads, measured by the angular shift (AS) of a CLIP-aligned translation when the null component is removed. We audit this ranking. The fraction of feature energy in the null space, ρ, reproduces SING's published order as well as SING's own score does (Kendall τ=0.82 for both; paired difference 95% CI $[-0.36, 0.29]$), and each keeps a partial association with the order given the other (τ=0.61 both ways). SING's ranking is therefore indistinguishable from a null-energy ranking. Our interventions show that AS has a path from null energy to the score that carries no image-specific content. They cannot, however, separate null energy from real null content as the source of the agreement. Replacing each model's null coordinates with content-free surrogates of the same energy reproduces SING's order (point τ 0.49–0.72 on all 13 models, depending on the surrogate), and so does the excess of real over surrogate AS (τ=0.62). The data cannot apportion the order between the two. Removing an equal share of null energy from every model amounts to measuring AS per unit null energy. With SING's released translators this loses the agreement (τ=−0.03 and 0.09 on two model sets; 0.61 on the second set without two models). With retrained ridge translators it keeps it (τ=0.61 and 0.75), at the level our earlier pre-registered rule set to reject the energy reading. Shuffling class means across classes in the null space, which preserves energy exactly, keeps the agreement (τ=0.64), but given ρ the class-aligned part has the stronger partial association (0.81 against 0.22). We read this as leaning toward energy; our pre-registration allows no stronger claim than these raw numbers. In semi-synthetic calibration worlds with known content, no estimator we tested meets all three criteria; the translator-free $\mathrm{NCM}_{\mathrm{null}}$ tracks content in two of the three worlds. Under our ridge translators, two true DINO backbones rank 8th and 9th of 15 by raw AS. Adjusted for null energy, ViT-S/16 falls below the prediction interval, which leans toward SING's account for that model, while ViT-B/16 lies inside it, so the evidence is mixed. SING's "DinoViT" translator matches a supervised timm ViT-L/16 checkpoint. Size-corrected scores change the order but none reverses it with confidence. The main analyses were pre-registered over four rounds (two round-4 addenda were written after the first round-4 results), post hoc analyses are labelled, and we report every pre-registered outcome. We recommend reporting ρ and an energy-only baseline next to any cross-model AS-type null-space score.

1 Introduction

A linear classifier W on features $f \in \mathbb{R}^m$ ignores every component of f in its null space $\ker W$. SING (Yadid et al., 2026) reads class-relevant content in that null space as "leakage": information the head could have used but did not. To measure it, SING maps features into CLIP space (Radford et al., 2021) with a learned linear translator, removes the null component, and records how far the translated feature turns relative to the class text (the absolute shift, AS). It uses AS and an IS/AS ratio to rank 13 ImageNet models and reports that a model labelled "DinoViT" (Caron et al., 2021) leaks least.

Null spaces matter well beyond SING. They appear in work on out-of-distribution failure (Idnani et al., 2023), outlier and OOD detection (Wang et al., 2022; Cook et al., 2020; Zöngür et al., 2025), overfitting (Rezaei & Sabokrou, 2023) and information hiding (Li & Short, 2024). Invariance-based adversarial examples show that class information can sit in directions a classifier ignores (Jacobsen et al., 2019). In NLP, nullspace projection and closed-form erasure remove a property from representations (Ravfogel et al., 2020; Elazar et al., 2021; Belrose et al., 2023). Linear maps between a vision model and CLIP, the kind of translator SING relies on, have been studied by Moayeri et al. (2023). Probing work has also shown that readouts need controls (Hewitt & Liang, 2019), and similarity scores are known to be dominated by high-variance directions (Davari et al., 2023).

Across architectures, the null dimension d_{null} and the null energy fraction

$$\rho = \mathbb{E}\|\Pi f\|^2 / \mathbb{E}\|f\|^2, \qquad \Pi = V_n V_n^\top,$$

vary by orders of magnitude (Table 8). Removing more energy should move a translated feature further whether or not that energy is semantic. If so, a cross-model comparison of null "semantics" would be tied to energy. This paper tests that possibility.

Contributions.

1. **Energy-only control** (§3). ρ alone reproduces SING's 13-model order as well as SING's score. The two are statistically indistinguishable predictors, and neither subsumes the other.
2. **Interventions** (§4). Equal-energy removal, six null-content surrogates, a class-mean permutation, retrained translators and three translator-free or energy-free scores. AS has a content-free energy path, but the data cannot apportion SING's order between null energy and null content, and the equal-energy result depends on the translator.
3. **Calibration** (§4.6). In semi-synthetic worlds with known null content, none of eight estimators meets all three calibration criteria, although NCM_{null} tracks content in two of the three worlds. A geometric decomposition describes how AS grows with energy (consistent with two held-out models).
4. **DINO checkpoints** (§5). SING's "DinoViT" translator fits a supervised timm ViT-L/16. Under our ridge translators, two true DINO backbones with linear heads fall mid-table by raw AS; adjusted for energy, the evidence is mixed.
5. **Size-corrected scores** (§6). They change the order, but no correction reverses it with confidence. Several class-level claims do not reproduce on our pipeline.
6. **Full reporting**. Four rounds of pre-registration with every outcome listed (Appendix A), including the kills of two score searches and the flaw in the flatness rule behind them (Appendix D).

2 Setup

SING's score. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space, W the model's head and V_n an orthonormal basis of $\ker W$ from the SVD of W (absolute singular-value tolerance 10^{-5} unless stated). For a text direction t, the per-image absolute shift for a removed component r is $|\theta(T(f), t) - \theta(T(f - r), t)|$. AS_{SING} is its mean over images with $r = \Pi f$ (full null removal) and the class text (the normalised mean of SING's six prompts) as t. We use SING's released code (commit `e47b37d`) and all 13 released translators on a fixed 10k-image ImageNet-val subset (10 images per class, all 1000 classes). SING's own model comparison also uses 10k features over the 1000 classes; its 16-class, 2000-image subset is the translator-validation set, which we do not use. For the class-level claims we add 200 images (all 50 validation images of jellyfish, porcupine, Arabian camel and sports car).

Reference order and statistics. SING's order is the 13 IS/AS bars of its Supp. Fig. 12b, which we digitised. A lower ratio means more leakage. Agreement is Kendall τ_b between a leakage score and that order. Every correlation with SING's order in this paper uses this sign convention: positive means that more of the score goes with more leakage (a lower IS/AS ratio). Model-level uncertainty comes from 2000

bootstrap resamples of models (seed 0), with percentile and BCa intervals (Efron, 1987). Paired comparisons of two scores use the same resamples. With n=13 these intervals are wide and often reach 1, so we read them as coarse. Sets: *all 13*; *E12*, the 12 models with $\rho \geq 0.005$ (all but DenseNet121, whose ρ=1.8×10^{-8}); and *S11*, a screened set declared in round 2 that also drops BiT. The screen removes models with $\rho < 10^{-4}$ or with a retained row-space direction of singular value $\sigma < 10^{-3}$ that holds more than 5% of the feature energy. BiT has such a direction (σ=8.9×10^{-4}, 26% of the energy). ResNet50 has one just under the threshold ($\sigma < 10^{-3}$, 4.4% of the energy): at tolerance 10^{-4} that direction joins the null space, ρ rises from 0.146 to 0.190 and $\mathrm{AS}_{\mathrm{SING}}$ falls from 1.68° to 1.33°.

Pre-registration. The main experiments below were declared in writing before their numbers were computed, over four rounds. Two round-4 addenda (R4B and R4C) were written after the first round-4 results were known, and R4C also after part of R4B and some exploratory geometry had run; each file lists the values known when it was written. The ledger is in Appendix A. Results marked *post hoc* or *exploratory* were added after seeing data. Commit times are set by the committer, so the repository cannot prove the order of events by itself. The hosting service's push history orders only commits that were pushed separately: the round-3 pre-registration was pushed on its own about 68 minutes before its results, and the round-4 pre-registrations were pushed before theirs.

3 Null energy reproduces SING's order

Reproduction. $\mathrm{AS}_{\mathrm{SING}}$ agrees with SING's order at τ=0.82 (95% CI $[0.56, 1.00]$). The like-for-like comparison, our IS/|AS| ratio against SING's IS/AS ratio, gives τ=0.744 (percentile $[0.33, 1.00]$, BCa $[0.12, 0.94]$). SING's ratios are 0.98–3.35 times ours (Table 1). The gap is largest for DinoViT (3.35×) and EVA02 (2.5×), both small-ρ models. SING's model comparison uses the same number of images and classes as ours, so the gap is not explained by the subset size. Its likely sources are SING's specific images, its concept set or its checkpoint (§5), and we cannot separate them without SING's evaluation lists.

ρ alone reproduces the order. Ranking models by ρ alone gives τ=0.82 $[0.56, 1.00]$, the same as $\mathrm{AS}_{\mathrm{SING}}$ (Fig. 1). On S11 the values are 0.78 for ρ and 0.82 for $\mathrm{AS}_{\mathrm{SING}}$. The paired bootstrap difference $\tau(\rho) - \tau(\mathrm{AS}_{\mathrm{SING}})$ is 0.00 with 95% CI $[-0.36, 0.29]$ (BCa $[-0.37, 0.28]$). Both agreements are far from chance (permutation p=5×10^{-5} for each, the smallest value 20000 permutations can give). ρ and $\mathrm{AS}_{\mathrm{SING}}$ agree with each other at τ=0.69 (Spearman 0.87). Each keeps a partial association with SING's order given the other: partial Kendall τ=0.61 both ways, and partial Spearman 0.63 for $\mathrm{AS}_{\mathrm{SING}}$ given ρ and 0.72 for ρ given $\mathrm{AS}_{\mathrm{SING}}$. The data therefore cannot tell the two predictors apart, and ρ does not make $\mathrm{AS}_{\mathrm{SING}}$ redundant. Null dimension alone is a weak predictor (Spearman +0.46 with the leakage order, p=0.12): DeiT3-L (Touvron et al., 2022) has d_{null}=24 yet the worst ratio. A NuSA-style null norm, $\mathbb{E}_i\|\Pi f_i\|/\|f_i\|$, in the spirit of Cook et al. (2020), is an energy proxy rather than a second leakage score: it orders the models almost as ρ does (τ=0.97) and agrees with SING at τ=0.85.

Within a model, AS grows with removed energy. On ResNet50, ResNeXt101, BiT-R50x1 (Kolesnikov et al., 2020) and DINOv2-L (Oquab et al., 2024), all with d_{null}=1048, removing random d-dimensional subspaces of the null space with $d \in \{24, \ldots, 1048\}$ raises AS about tenfold from d=24 to d=1048 on three of four models (Table 16, Fig. 3 in Appendix C). For random subspaces, removed energy is close to linear in d, so this experiment cannot separate energy from dimension. A second experiment mixes high- and low-variance null PCs to vary energy at fixed d and dimension at fixed energy. It fits $\log|\mathrm{AS}| = a\log\rho + b\log d + c_{\mathrm{model}}$ over 76 cells on five models with a=0.431 (cell bootstrap $[0.388, 0.485]$, model-clustered bootstrap $[0.382, 0.464]$, leave-one-model-out 0.408–0.452; the clustered CI rests on five clusters, one with a single cell, so it is not more reliable for being narrower) and a dimension term b=0.010 whose CI covers 0 (Appendix C). A first-order calculation gives a=0.5: if the translator maps the removed component linearly and the residual is small, the angle changes in proportion to $\|\Pi f\| \propto \sqrt{\rho}$. The pre-registered flatness part of this experiment failed (max/min across d at fixed ρ up to 2.29 against a 1.5 bar), so we claim only $a \approx 0.4$–0.5.

Model	ρ	ours IS/|AS|	SING IS/AS	SING/ours
VGG16	0.350	17.75	24.26	1.37
VGG19	0.339	15.56	19.89	1.28
DenseNet121	1.8×10^{-8}	41.48	70.71	1.70
ResNet50	0.146	17.40	24.81	1.43
"DinoViT"	0.005	20.95	70.16	3.35
EffNet-B0	0.130	22.15	36.28	1.64
BiT-R50x1	0.046	24.93	47.21	1.89
ResNeXt101	0.451	12.60	13.88	1.10
ConvNeXt-B	0.013	32.86	37.38	1.14
Swin-L	0.274	21.21	29.18	1.38
DINOv2-L	0.322	15.49	16.61	1.07
DeiT3-L	0.348	9.73	9.51	0.98
EVA02-L	0.009	21.64	54.32	2.51

Table 1: Like-for-like reproduction. Our IS/|AS| on the 10k subset against SING's IS/AS (digitised Supp. Fig. 12b). $\tau(\text{ours}, \text{SING})$=0.744 [0.33, 1.00].

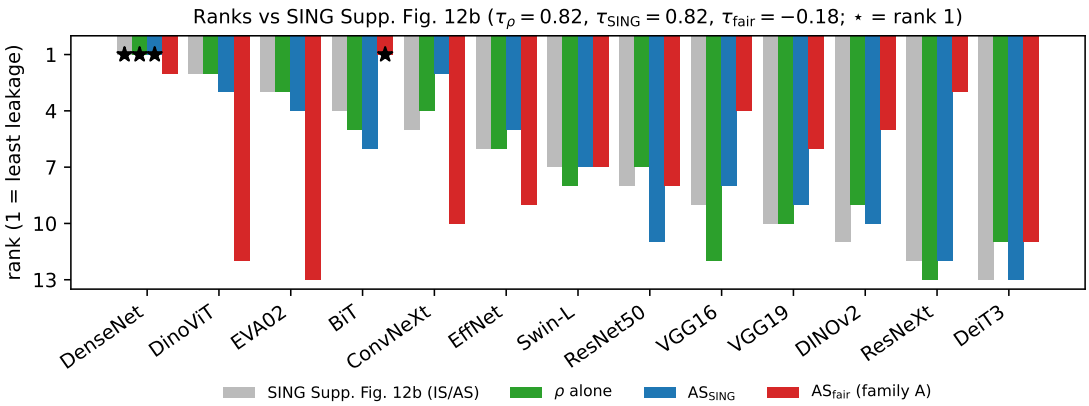


Figure 1: Ranks (1 = least leakage, ⋆ marks rank 1) under SING Supp. Fig. 12b, ρ alone, AS_{SING} and the size-corrected AS_{fair} (family A, §6).

BiT is the exception. Its full-null AS collapses (r=0.06) while its partial subspaces give large AS (9–13°), because the translator's null blocks nearly cancel (top singular value of AV_n 284, against 5.6 for ResNet50). This is also why BiT fails the screen.

4 Energy or content? Interventions

Agreement between ρ and SING's order does not show that energy drives AS across models. Energy could instead *mediate* genuine leakage, if models with more null energy also hold more class content there. We ran four interventions in round 3 and further tests in round 4, pre-registered unless marked post hoc or exploratory (Tables 3 and 2, Fig. 2). None of them separates the two readings. They do show that AS has a content-free path from null energy to the score. They also show that the data cannot apportion SING's order between null energy and null content. We read a τ of 0.6 or more as tracking and $|\tau| < 0.3$ as not reproducing the order; with 13 models, most CIs are wide, so we treat point estimates between the bars as inconclusive. In round 3 the bars were applied asymmetrically: $\mathrm{AS}_{\mathrm{surr}}$ was read on its point $\tau \geq 0.6$, while X needed its whole CI above 0.6.

4.1 Equal null energy (released translators)

Design. In *A1* we remove $\beta\Pi f$ with $\beta = \sqrt{\rho_0/\rho}$, ρ_0=0.005, so every model loses the same fraction of its feature energy, taken from its own null space in its own proportions. *A2* removes the same energy from the top-variance null PCs, and *A3* removes it from a random rotation of the null basis. *A4* matches energy in CLIP space instead: with $\kappa = \mathbb{E}\|A\Pi f\|^2/\mathbb{E}\|Tf\|^2$, we remove $\beta'\Pi f$ with $\beta'^2\kappa = \kappa_0$=0.0023. The pre-registered outcome was ENERGY-ACCOUNTS if $\tau(\mathrm{A1}) < 0.3$ and the paired CI of $\tau(\mathrm{AS}_{\mathrm{SING}}) - \tau(\mathrm{A1})$ is above 0 on both E12 and S11, and a kill of the energy reading if $\tau(\mathrm{A1}) \geq 0.6$ on both.

What A1 measures. For a small removal, the translated shift is close to linear in the removed component, so A1 $\approx \beta\,\mathrm{AS}_{\mathrm{SING}} = \mathrm{AS}_{\mathrm{SING}}\sqrt{\rho_0/\rho}$: A1 is, to first order, AS per unit null amplitude. The data agree: A1 and $\mathrm{AS}_{\mathrm{SING}}/\sqrt{\rho}$ give nearly the same order (τ=0.79 on E12 and 0.82 on S11; Spearman 0.92 and 0.93), and A1 and $\mathrm{AS}_{\mathrm{SING}}/\rho^{\hat{a}}$ give τ=0.73 and 0.75. The first-order ratio $\mathrm{A1}/(\beta\mathrm{AS}_{\mathrm{SING}})$ runs from 0.36 (DeiT3-L) to 1.00 (DinoViT). The question A1 answers is therefore whether AS *per unit null energy* tracks SING's order. Whether energy causes AS is a separate question.

Result. With the released translators, AS per unit energy does not track SING: τ=-0.03 $[-0.54, 0.61]$ on E12 and 0.09 $[-0.45, 0.78]$ on S11. The paired drop from $\mathrm{AS}_{\mathrm{SING}}$ has CI $[0.23, 1.39]$ (BCa $[0.25, 1.41]$) on E12 and $[0.08, 1.30]$ on S11, which meets the pre-registered ENERGY-ACCOUNTS criterion. Taking the energy from top PCs (0.03) or random directions (0.03) gives the same picture. The result rests on few models. Delete-one jackknife values of τ(A1) span $[-0.16, 0.13]$ on E12 and $[-0.02, 0.29]$ on S11, but dropping DinoViT and EVA02 together gives 0.38 on E12 and 0.61 on S11. The S11 value reaches the level (≥ 0.6) that our round-3 kill rule set for the energy reading, and no other of the 55 leave-two-out subsets of S11 does. Matching CLIP-space energy instead (A4: -0.15 and -0.20) gives the same result, but only with the released translators. With our ridge translators A4 gives 0.52 and 0.49 (§4.4). We therefore do not use A1 or A4 as evidence about the cause of the order. The A1 scores still differ across models by a factor of six (Table 8), so equalising energy changes their order without making them equal.

Translator gain. CLIP-space energy κ tracks SING's order slightly less well than ρ (τ=0.67 on E12). The paired difference $\tau(\kappa) - \tau(\rho)$ has CI $[-0.41, 0.28]$, so we keep ρ as the named variable. The translator's null gain $\gamma = \kappa/\rho$ does not track SING (τ=-0.06). Differences in translator gain γ therefore do not explain the order, although gain does shape the A1 values themselves (§4.4).

4.2 Leak-free null content

Design. We replace each image's null coordinates $c_i = V_n^\top f_i$ with a draw $\tilde{c}_i \sim \mathcal{N}(\mu, \Sigma)$, where μ and Σ are the mean and covariance of c over the 10k images (5 draws). The row-space part, and so every logit, is unchanged. The surrogate carries no image-specific and thus no class-specific content. On ResNet50, ResNeXt101 and DINOv2-L we scale the surrogate to null energy fractions $\rho_t \in \{0.01, 0.05, 0.2, 0.45\}$. We then optionally plant a concept direction with strength $\alpha \in \{0, 0.1, 0.3\}$, using the generator of our holdout test (Appendix D; concepts water, green, thin, knitted). The pre-registered test required that, at α=0, AS rise with ρ_t with Spearman $\geq$0.8 and that the log-log slope CI exclude 0, on all three models.

Score	E12	S11
A1 and its influence (released)		
A1, all models	-0.03 $[-0.54, 0.61]$	0.09 $[-0.45, 0.78]$
A1 without DinoViT and EVA02	0.38	0.61
Retrained ridge translators		
A1	0.61 $[0.15, 0.94]$	0.75 $[0.32, 1.00]$
A4 (matched CLIP-space energy)	0.52 $[-0.10, 0.97]$	0.49 $[-0.17, 1.00]$
κ	0.76 $[0.40, 0.97]$	0.75 $[0.32, 1.00]$
Class-mean permutation (idea 1)		
$\mathrm{AS}_{\mathrm{cperm}}$, released	0.64 $[0.28, 0.90]$	0.64 $[0.22, 0.92]$
Δ_{align}, released	0.64 $[0.28, 0.90]$	0.64 $[0.24, 0.91]$
$\mathrm{AS}_{\mathrm{cperm}}$, ridge	0.70 $[0.33, 0.94]$	0.67 $[0.24, 1.00]$
Δ_{align}, ridge	0.09 $[-0.57, 0.61]$	0.13 $[-0.58, 0.63]$
Translator-free and energy-free scores		
$\mathrm{NCM}_{\mathrm{null}}$ (idea 5)	0.45 $[-0.11, 0.90]$	0.42 $[-0.22, 0.92]$
null zero-shot NZS (B3)	0.30 $[-0.31, 0.89]$	0.38 $[-0.28, 0.96]$
selectivity R, k=20 (idea 3)	0.15 $[-0.44, 0.59]$	0.05 $[-0.59, 0.63]$
Geometry (exploratory)		
$\mathrm{AS}_{\mathrm{oop}}$ (out-of-plane only)	0.67 $[0.37, 0.93]$	0.67 $[0.35, 0.96]$
$\mathrm{AS}_{\mathrm{in}}$ (in-plane only)	0.67 $[0.33, 0.93]$	0.67 $[0.29, 1.00]$
SING's IS (B4)		
IS against row-space energy $1-\rho$	-0.55 $[-0.83, -0.10]$	-0.49 $[-0.84, 0.02]$

Table 2: Round-4 scores against SING's order (Kendall τ, percentile 95% CI, 2000 model resamples). A1: AS after removing $\sqrt{\rho_0/\rho}\,\Pi f$ (§4.1); ridge: our retrained translators on the test half (§4.4); $\mathrm{AS}_{\mathrm{cperm}}$ and Δ_{align}: §4.3; $\mathrm{NCM}_{\mathrm{null}}$, NZS and R: §4.6; geometry and IS: §4.7.

Score	all 13	E12	S11
ρ (null energy fraction)	0.82 $[0.56, 1.00]$	0.79 $[0.49, 0.97]$	0.78 $[0.44, 1.00]$
$\mathrm{AS}_{\mathrm{SING}}$ (released translators)	0.82 $[0.56, 1.00]$	0.79 $[0.51, 1.00]$	0.82 $[0.51, 1.00]$
κ (CLIP-space null energy)	0.72 $[0.44, 0.95]$	0.67 $[0.37, 0.93]$	0.67 $[0.35, 0.96]$
NuSA-style null norm	0.85 $[0.64, 1.00]$	0.82 $[0.57, 0.97]$	0.82 $[0.53, 1.00]$
$\gamma = \kappa/\rho$ (translator null gain)	-0.10 $[-0.56, 0.43]$	-0.06 $[-0.59, 0.54]$	0.05 $[-0.51, 0.74]$
Leak-free surrogate $\mathrm{AS}_{\mathrm{surr}}$	0.69 $[0.39, 0.92]$	–	0.64 $[0.27, 0.92]$
Equal-energy interventions (released translators)			
A1: remove $\beta\Pi f$, $\beta^2\rho = 0.005$	–	-0.03 $[-0.54, 0.61]$	0.09 $[-0.45, 0.78]$
A1′: same, $\rho_0 = 0.001$	–	-0.03 $[-0.54, 0.61]$	0.09 $[-0.45, 0.78]$
A2: top null PCs, energy ρ_0	–	0.03 $[-0.54, 0.54]$	0.13 $[-0.43, 0.67]$
A3: random null rotation, energy ρ_0	–	0.03 $[-0.47, 0.67]$	0.16 $[-0.50, 0.84]$
A4: matched CLIP-space energy κ_0	–	-0.15 $[-0.68, 0.44]$	-0.20 $[-0.74, 0.46]$
Retrained ridge translators (test half)			
$\mathrm{AS}_{\mathrm{SING}}$	0.85 $[0.61, 1.00]$	0.82 $[0.57, 1.00]$	0.82 $[0.51, 1.00]$
A1	–	0.61 $[0.15, 0.94]$	0.75 $[0.32, 1.00]$
A1, λ on extended grid†	0.49 $[-0.10, 0.91]$	0.39 $[-0.21, 0.86]$	0.53 $[-0.06, 1.00]$

Table 3: Kendall τ against SING's order (percentile 95% CI, 2000 model resamples). E12 drops DenseNet ($\rho < \rho_0$); S11 also drops BiT. A1′ gives the same order as A1 because, for each model, AS is monotone in the removed fraction. †Post hoc: ridge penalty chosen on an extended grid $10^{3..6}$, since the pre-registered grid put every model at its upper edge.

Result. The test passes on all three models (Spearman 1; slopes 0.38–0.69, all CIs far from 0; Table 9 in Appendix B). AS therefore grows with null energy that carries no image-specific content, which gives it a content-free path from null energy to the score.

Planted content. Planted content raises AS at fixed energy, by ×1.04–1.09 at α=0.1 and ×1.35–1.51 at α=0.3 (ρ_t=0.2), and by ×1.14–1.58 at α=0.3 when each model keeps its own ρ (Table 13 in Appendix B; DINOv2-L ×1.17). Across the interquartile range of ρ over the 12 models (0.038 to 0.341), the energy law

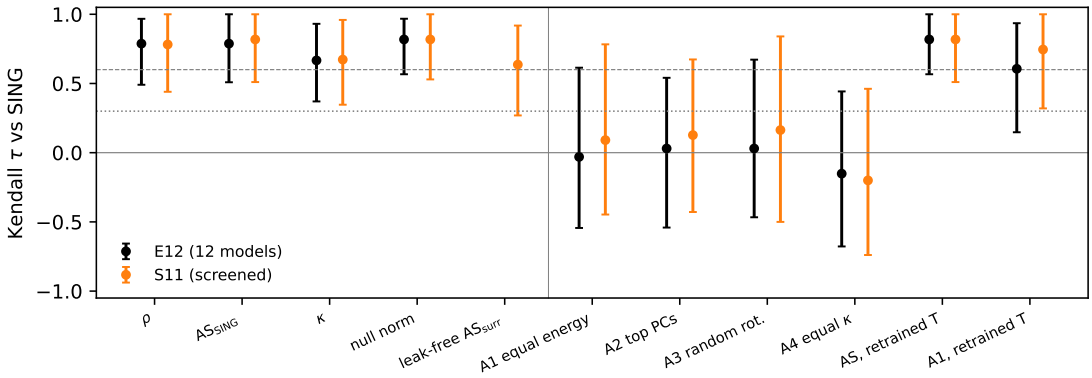


Figure 2: Agreement with SING's order under the energy controls and interventions of §4 (95% percentile CIs). Dotted line: τ=0.3; dashed: τ=0.6, the pre-registered bars.

moves AS by r_E=2.4–5.4× on these three models. The planted strength that would move AS as much, α^*, is therefore above 0.3, the largest strength we tested, on all three models. So AS does respond to null content at fixed energy. At ρ_t=0.2 and at each model's own ρ, though, content moves it less than the observed spread of energy does. This does not hold at the lowest energy (post hoc, from the same sweep): at ρ_t=0.01, α=0.3 moves AS by ×2.70 (ResNet50), ×3.47 (ResNeXt101) and ×5.26 (DINOv2-L), above r_E (2.39, 5.37 and 4.92) on 2 of the three models.

Leak-free scores and the content excess. On all 13 models, replacing each model's null content with its own leak-free surrogate, at its own energy and covariance, gives $\mathrm{AS}_{\mathrm{surr}}$ with τ=0.69 [0.39, 0.92] against SING (0.64 on S11) and Spearman 0.96 with $\mathrm{AS}_{\mathrm{SING}}$. In round 4 we pre-registered both halves of this decomposition and five more surrogates (Table 12 in Appendix B).

- *Content-free half.* AS with surrogate null content tracks SING ($\tau \geq 0.6$) for three of the five content-free surrogates. Its value depends on the surrogate: point τ on all 13 models runs from 0.49 (isotropic and mean-only) to 0.72 (zero-mean Gaussian), and $\mathrm{AS}_{\mathrm{surr}}$ with the same Gaussian tracks CLIP-space null energy κ (τ=0.92). Real null coordinates permuted across images ($\mathrm{AS}_{\mathrm{perm}}$) give 0.69 [0.39, 0.92] on all 13 and 0.64 on S11.
- *Content half.* The excess $X = \mathrm{AS}_{\mathrm{SING}}/\mathrm{AS}_{\mathrm{surr}}$ has τ=0.54 [0.14, 0.83] on all 13 and 0.60 [0.24, 0.91] on S11, and $\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{perm}}$ has 0.62 [0.15, 0.92] and 0.71 [0.28, 1.00]. Given ρ, the partial Spearman of X with the order is 0.79 [0.02, 0.97] on S11, against 0.52 for $\mathrm{AS}_{\mathrm{perm}}$.

Both halves track the order, the outcome the pre-registration calls "cannot apportion". The data cannot apportion SING's order between null energy and image-specific null content.

4.3 Class-mean permutation

Design. Write each image's null coordinates as $z_i = \mu_{y_i} + r_i$, with μ_c the class mean over its 10 images and r_i the within-class residual. We draw a derangement π of the 1000 classes (5 draws) and set $z'_i = \mu_{\pi(y_i)} + r_i$. The logits are unchanged, and because every class has 10 images, the total null energy and the uncentred null second-moment matrix are preserved exactly (largest relative error 2×10^{-16}). Only the alignment between an image's class and its class-mean null content is broken. $\mathrm{AS}_{\mathrm{cperm}}$ is the mean AS over draws, $\Delta_{\mathrm{align}} = \mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{cperm}}$ and the share is $\Delta_{\mathrm{align}}/\mathrm{AS}_{\mathrm{SING}}$.

Result. With the released translators, $\mathrm{AS}_{\mathrm{cperm}}$ still tracks SING (τ=0.69 [0.34, 0.94] on all 13, 0.64 on E12, 0.64 on S11), and the median share is 0.10 (E12) and 0.14 (S11). With our ridge translators the pattern is the same ($\mathrm{AS}_{\mathrm{cperm}}$ 0.70 and 0.67; median share −0.04). This is the pre-registered outcome ENERGY-CARRIES under both translator families. The outcome was largely foreseeable once $\mathrm{AS}_{\mathrm{perm}}$ was known to track the

order, since both surrogates preserve null energy. Δ_{align} itself tracks the order with the released translators (0.64, 0.64) but not with ours (0.09, 0.13).

Reading. We take this as evidence leaning toward energy, not as a demonstration, for two reasons. First, given ρ, the class-aligned part keeps the stronger association with the order: with the released translators the partial Spearman is 0.22 (all 13), 0.22 (E12) and 0.35 (S11) for AS_{cperm}, against 0.76, 0.81 and 0.88 for Δ_{align}. Second, Δ_{align} fails the calibration of §4.6: at fixed class content it falls as energy rises (τ=−1.00 in world C on all three bases). Our pre-registration allows no stronger claim from this test than its raw numbers.

4.4 Retrained translators

SING's translators were fit by its authors, and AS depends on them. We fit our own ridge translators (Hastie et al., 2009, §3.4) $f \mapsto$ CLIP image embedding on half of the 10k images (5000; penalty chosen on a held-out split of that half) and evaluate on the other half. Test fidelity (mean cosine) is 0.79–0.84, against 0.81–0.87 for the released translators on the same images. SING does not list the images its translators were fit on, so on our subset their fidelity may be partly in-sample; ours is measured out of sample. With retrained translators, AS_{SING} still agrees with SING (τ=0.85 $[0.61, 1.00]$; 0.82 on S11; τ=0.72 against released-translator AS). The AS order is thus not specific to the released translators.

The equal-energy result is. With retrained translators, A1 gives τ=0.61 $[0.15, 0.94]$ on E12 and 0.75 $[0.32, 1.00]$ on S11. Read against the round-3 rule (a kill of the energy reading if $\tau(\text{A1}) \geq 0.6$ on both E12 and S11), this is at the kill level. The rule was registered for the released translators, so we apply it as a reading, not as a registered outcome. The paired CI of $\tau(\text{AS}_{\text{SING}}) - \tau(\text{A1})$ covers 0 ($[-0.20, 0.65]$ on E12, $[-0.32, 0.49]$ on S11). Leaving out any two models keeps it at 0.42 or more on E12 and 0.61 or more on S11. Released and retrained A1 order the models almost independently (τ=0.06 on E12). Retrained A1 spans a narrow range (0.048–0.091°, against 0.049–0.302° released) and follows ρ (τ=0.58 and 0.67). Given ρ, its partial Spearman with the order is 0.15 (E12) and 0.41 (S11). With these translators, matched CLIP-space energy (A4) gives 0.52 $[-0.10, 0.97]$ and 0.49, between our bars, and κ itself tracks the order (0.76, 0.75). The pre-registered penalty grid put every model at its upper edge, so we extended the grid post hoc (10^3–10^6). Over the combined grid the chosen penalties are interior: 10^4 for VGG16, VGG19 and DenseNet, and 10^3, the value already used, for the other ten models. With these penalties A1 gives 0.39 $[-0.21, 0.86]$ on E12 and 0.53 on S11. The drop from 0.61 comes from re-fitting just the two VGG models, so this number is fragile. Forcing heavier regularisation on every model (10^6) raises it to about 0.79. Whether AS per unit energy tracks SING's order therefore depends on how the translator is fit. It does not with the released translators, and it does with ours. We do not know which translator better reflects null semantics, so neither result decides between the two readings.

4.5 Are high-energy null directions more semantic?

If the high-variance null directions were the semantic ones, energy would partly *be* the signal. At matched d, we compare the top-variance null PCs with random null subspaces. In round 2 we used $G = \text{AS}/\sqrt{\rho}$. The pre-registered round-2 label was "top PCs carry less AS per unit energy" on 2 of the 3 screened models at d=24: $G_{\text{top}}/G_{\text{rand}}$ was 0.60 for ResNeXt101 and 0.87 for DINOv2-L, with CIs below 1, against 1.16 for ResNet50. At d=100 the ratios were 1.01 (ResNet50), 0.85 (ResNeXt101), 0.83 (DINOv2-L) and 0.011 (BiT, not counted). However, $\sqrt{\rho}$ is not the fitted law. In round 3 we compare against $\rho^{\hat{a}}$, $H = (\text{AS}_{\text{top}}/\rho^{\hat{a}}_{\text{top}})/(\text{AS}_{\text{rand}}/\rho^{\hat{a}}_{\text{rand}})$. At $\hat{a}$=0.431 and d=24, H is 1.50 (ResNet50), 0.74 (ResNeXt101) and 0.98 $[0.97, 0.99]$ (DINOv2-L), which gives the pre-registered label "top PCs carry less AS than energy predicts" (2/3). DINOv2-L's H differs from 1 by 2%, a negligible effect even though its CI excludes 1. At the low end of the CI for a (0.388), the label flips to "more", so it is *not robust* (Table 10 in Appendix B). High-energy null directions are not clearly more semantic per unit energy. The result varies across models and with the exponent, so this test does not decide between the readings.

4.6 Calibration and translator-free scores

Semi-synthetic worlds. To check whether any estimator can tell null content from null energy, we built three worlds on each of three bases (ResNet50, ResNeXt101 and DINOv2-L, each with its released translator) by rescaling the class-mean part μ_y and the residual r of the null coordinates, with the 12 E12 values of ρ as targets. Content is measured by L, the class-mean null energy as a share of total feature energy. In world C, L is fixed while ρ varies; in world M the residual is fixed, so L rises with ρ; in world D, ρ is fixed and L varies. An estimator is *content-diagnostic* if, on all three bases, $|\tau(\cdot, \rho)| \leq 0.2$ in world C, $\tau(\cdot, L) \geq 0.8$ in world M and $\tau(\cdot, L) \geq 0.6$ in world D. None of the eight estimators we tested meets all three criteria (ρ, AS, A1, $\mathrm{AS}_{\mathrm{gauss}}$, X, Δ_{align}, $\mathrm{NCM}_{\mathrm{null}}$ and the selectivity R; Table 14 in Appendix B). The bar is strict. In world C, where L is fixed and ρ grows (the absolute class-mean null energy rises $\times 1.81$, since the row-space energy is held fixed), the class-mean share of null energy falls from 50% to 0.6%, and a score that responds to that share fails world C by construction. $\mathrm{NCM}_{\mathrm{null}}$ tracks L in worlds M (0.80–0.92) and D (0.93–1.00), and Δ_{align} and X track it in world D (0.94–1.00 for Δ_{align}); all three fail world C. AS follows ρ at fixed content (world C, τ=1 on all bases). The energy exponents in worlds C and M run from 0.28 to 0.55, against $\hat{a}$=0.431 from the energy law and 0.53 $[0.24, 0.82]$ across models.

Translator-free and energy-free scores.

- *Null decodability* $\mathrm{NCM}_{\mathrm{null}}$: leave-one-out nearest-class-mean accuracy on the normalised null coordinates, minus a shuffled-label control. It needs no translator. It ranges from 5.2 to 80.9 points and gives τ=0.45 $[-0.11, 0.90]$ on E12 and 0.42 on S11, between our bars. It follows ρ (0.48, 0.42), and its partial association given ρ is about zero (-0.05, -0.01).
- *Null zero-shot accuracy* NZS: zero-shot classification from the translated null part alone, minus a permuted control. It passes its validity gate (no change under rescaling of Πf; τ=1 with planted content on all three bases), but it is not energy-free ($\tau(\cdot, \rho)$=0.45 and 0.53), and against SING it gives 0.30 $[-0.31, 0.89]$ (E12) and 0.38 (S11): inconclusive. Given ρ, its partial association with the order has negative point estimates, but every CI includes 0 (-0.60 $[-0.89, 0.50]$, -0.63 $[-0.92, 0.47]$ and -0.54 $[-0.93, 0.86]$ on all 13, E12 and S11).
- *Selectivity* R: AS after removing the top-20 between-class null directions (estimated on held-out images) over AS after removing the same energy at random. Within each model R differs from 1 (CI excludes 1 on 12 of 12 models), but in both directions (4 above 1, 8 below). Across models it does not reproduce the order (τ=0.15 and 0.05; k=5 and 50 give the same).

None of these scores tracks SING's order. None is both energy-free and fully calibrated, so we propose no replacement score.

4.7 How AS grows with energy

Decomposition. Split the removed translated null component $\delta = A\Pi f$ into its part in the plane of $T(f)$ and the class text t and the orthogonal, out-of-plane part. The out-of-plane part changes the angle to t only at second order, by $\frac{1}{2} \cot\theta \, \|\delta_{\mathrm{oop}}\|^2 / \|T(f)\|^2$. It does so whatever direction δ_{oop} has, so it is blind to the class text. If the translator's squared gain per null PC scales with the PC's variance as λ^s, a first-order argument predicts the energy exponent $a_{\mathrm{pred}} = (1 + s)/2$.

Pre-registered tests. On a synthetic model with a known gain slope $s \in \{-0.4, \ldots, 0.2\}$ the fitted exponent is within 0.027 of $(1 + s)/2$ (G3, pass). On the two true DINO backbones of §5, which were not used to design the test, the pre-registered checks pass on both: the Gaussian closed form orders the surrogate AS perfectly (Spearman 1), the out-of-plane-only removal shifts the angle ($[0.265, 0.269]$ and $[0.516, 0.523]$ degrees), the closed form is within 0.033 and 0.066 of the measured out-of-plane shift, and a_{pred} is within 0.1 of a_{fit} (0.603 against 0.669; 0.569 against 0.580; Table 15 in Appendix B). The out-of-plane share of AS is 0.52 and 0.80. These results are consistent with the account rather than a strong test of it: the Spearman check in G2 uses only four ρ_t points, and the G3 generator builds in the gain law. This gives a mechanism

by which AS responds to null energy without regard to the class text. It does not show that the null space holds no semantics.

Exploratory, 13 models (post hoc). With the released translators the out-of-plane share of AS runs from 0.20 to 0.98, and the in-plane and out-of-plane parts track SING equally (τ=0.67 and 0.67 on E12). a_{pred} is within 0.1 of a_{fit} on 6 of the 8 models where both can be computed; it misses on BiT (0.08 against 0.40) and on DINOv2-L (0.29 against 0.39).

SING's IS. SING's invariant-space shift IS is not a proxy for row-space energy: $\tau(\text{IS}, 1-\rho)$ is -0.62, -0.55 and -0.49 on the three sets, short of the pre-registered 0.6 on two. IS does, however, rise with null energy (CIs $[-0.89, -0.21]$, $[-0.83, -0.10]$ and $[-0.84, 0.02]$, excluding 0 on two sets), and it tracks SING's order (τ=0.74, 0.70 and 0.71).

4.8 What the interventions show

Energy and SING's score cannot be separated by their agreement with SING's order (§3). AS has a content-free energy path. It grows with surrogate null content that has no image-specific content, and the out-of-plane mechanism explains how. Content-free surrogates at each model's energy reproduce the order, and so does the content excess. The data cannot apportion the order between the two. AS per unit energy loses the agreement with the released translators and keeps it with ours. Shuffling class means keeps the agreement (τ=0.64 on E12), which leans toward energy, but given ρ the class-aligned part has the stronger association (0.81 against 0.22), and our pre-registration allows no stronger claim than these raw numbers. No estimator we tested meets all three calibration criteria, so this cannot settle the question. AS responds to planted content at fixed energy, so these results do not show that AS carries no leakage signal.

5 The DinoViT checkpoint

SING reports its "DinoViT" at 84.0% top-1 and itself describes the model as pretrained beyond ImageNet. Its released configuration loads a timm AugReg ViT-L/16. We evaluated five public ViT-L/16 checkpoints on the full 50k validation set: torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear (Singh et al., 2022) (85.14%), SWAG end-to-end (88.07%) and `google/vit-large-patch16-224` (81.98%) (Fig. 7, Appendix C). None falls within 84.0 ± 0.3. SING's evaluation protocol is not specified, so the 84.0% versus 85.85% gap is unexplained.

With SING's `dinovit1` translator on 1000 images, the timm AugReg features give mean cosine 0.848 to the CLIP image embeddings. Their shuffled-pair floor is 0.575, centred cosine is 0.674 and retrieval top-1 is 78.8%. Every other candidate, re-extracted with its native transform, is at its own shuffled floor (largest gap 0.002) with retrieval top-1 at most 0.2%. SING's translator was thus fit on this supervised timm checkpoint, while the other candidates are at chance. DINO never released a ViT-L with a 1000-way head (Caron et al., 2021). SING's narrative about self-supervision should therefore not rest on this model.

True DINO backbones. In round 4 we evaluated the two DINO backbones that were released with a linear ImageNet head, ViT-B/16 and ViT-S/16 (Caron et al., 2021), together with a supervised timm AugReg ViT-B/16 as a control (Table 4). Their 50k top-1 is 78.06%, 76.91% and 84.54% (the DINO values are within 0.2 points of the published linear-evaluation accuracies), and the pre-registered feature checks pass. Each DINO head leaves a null space of dimension 536 (ρ=0.189 and 0.437); the AugReg head has none. There is no released SING translator for these models, so we use ridge translators fit as in §4.4 (fidelity 0.819 and 0.816) and compare with the 13 models under their ridge translators. AS is 0.514° (ViT-B/16) and 0.652° (ViT-S/16), ranks 8 and 9 of 15 (median of the 13: 0.399°). The pre-registered raw-AS question (Q-DN1) asks whether both DINO models rank among the four lowest; its outcome is "SING-inconsistent", since both lie above the median of the 13. Raw ridge AS is not SING's own measure, however, and it tracks null energy, which for both models is above the median of the 13 (ρ=0.189 and 0.437), so, post hoc, we put more weight on the energy-adjusted reading (Q-DN2). Against the energy law fitted on E12, ViT-B/16 lies inside the 95% prediction interval ($[0.38, 0.88]$, predicted 0.58) and ViT-S/16 lies just below it ($[0.66, 1.57]$,

Model	top-1 (50k)	d_{null}	ρ	fidelity	AS (ridge) [rank/15]	95% PI from E12 law	NCM_{null}
DINO ViT-B/16	78.06	536	0.189	0.819	0.514 [8]	$[0.380, 0.884]$	56.9
DINO ViT-S/16	76.91	536	0.437	0.816	0.652 [9]	$[0.661, 1.573]$ (below)	53.4
AugReg ViT-B/16	84.54	0	0	–	–	–	–

Table 4: True DINO backbones with their released linear heads (§5), ridge translators on the test half. Rank among 15 models (1 = least leakage); prediction interval from the E12 energy law $\log \text{AS} \sim \log \rho$. AugReg ViT-B/16: supervised control with an empty null space.

predicted 1.02). ViT-S/16 thus has less AS than its energy predicts, which leans toward SING's account for that model, and the pre-registered outcome is mixed. For Δ_{align} and NCM_{null}, both models fall inside the prediction intervals from ρ. The pre-registered "least leaking" label, which needs both models among the four lowest, holds for neither score; ViT-S/16 alone is among the four lowest on both (Δ_{align} rank 3 of 15, NCM_{null} rank 2 of 14), while ViT-B/16 ranks 13 and 6. Both DINO heads are linear probes on frozen, concatenated block features, a different regime from the supervised heads of the 13 models, so this comparison is indicative only.

6 Size-corrected scores and SING's claims

Size-corrected scores. $\text{AS}_{\text{fair}} = \text{AS}_{\text{null}}/\text{AS}_{\text{rand}}$ divides AS by the AS of an energy-matched random removal. We used three families of controls, and name the family each time. In *family A*, the random subspace is Haar over the full feature space and changes the logits. *Family B* draws from the lowest-σ row-space directions of W. It is not logit-preserving either: it moves the logits by up to 14–74 logit SDs on models with a large span (VGG, ResNeXt; Table 8). *Family C* compares a 24-dimensional null subspace with a family-B control of matched energy. Point τ against SING on all 13 models is -0.18 (A), -0.41 (B) and -0.13 (C), and on S11 it is -0.56, -0.60 and -0.27. All fall below the pre-registered 0.3 bar.

Over eight normalisers (AS/ρ^a for $a \in \{0.25, 0.43, 0.5, 1\}$, $\text{AS}/d_{\text{null}}$, families A–C), the 95% CI excludes AS_{SING}'s 0.82 for 7 of 8 normalisers on all 13 models. On S11 it does so for exactly 5 of 8, so the weak "differs from" criterion passes at the threshold (5/8). The pre-registered reversal criterion (CI entirely below 0 under at least 5 of 8 normalisers) fails. It fails on all 13 models (0/8), and on S11 only family B meets it ($[-0.89, -0.23]$). That single pass comes from a control that is not logit-preserving. Normalising by the fitted exponent leaves a positive τ ($\rho^{0.43}$: $+0.33$ $[-0.15, 0.77]$; $\sqrt{\rho}$: $+0.26$), and family-A AS_{fair} over-corrects: its Spearman correlation with ρ is -0.79 $[-1, -0.23]$ on S11 (Fig. 5, Appendix C). Size-corrected rankings therefore *differ* from SING's, but the direction depends on the normaliser, and we make no reversal claim. Model ranks from AS_{SING} and AS_{fair} also shift with the null tolerance. Adjacent-tolerance τ falls to 0.87 and 0.82, below the pre-registered 0.9, while ρ stays at or above 0.92.

Class-level claims. For each of SING's testable claims we define a signed effect (positive = SING's direction) and bootstrap images within class (1000×, BCa intervals; Table 5). A claim *holds* if the CI is above 0, is *reversed* (rev.) if below, and is *not confirmed* (n.c.) if it covers 0. Rows that share a computation are merged. Under AS_{SING}, 7 claims hold, 3 reverse (one borderline) and 1 is contradicted. The reversed claims are C4 (DinoViT does not have the lowest class AS; ConvNeXt and DenseNet are lower), C8 (ResNet's porcupine and sports-car classes are not in its top decile) and C10 (jellyfish $\gg$ camel; effect -0.107, BCa $[-0.210, -0.006]$). C10 is borderline: its percentile interval $[-0.206, 7.9\times10^{-4}]$ covers 0. C9 ("no correlation" between ResNet and DinoViT class AS) is contradicted by a small correlation (-0.033, BCa CI excludes 0). These differences may reflect protocol differences (for C10, SING used about 30 concepts and 1300 images). They are not caused by size correction. Under AS_{fair} (family A) the other testable effects fall on the opposite side (opp.). Because AS_{fair} over-corrects, we treat these as diagnostics, not reversals. The image-bootstrap CIs ignore uncertainty from the choice of models and normaliser, so they are narrow. For C6 (class SD) the bootstrap now resamples classes, since the effect is a spread over classes; the point estimate lies inside both the BCa and the percentile interval under both scores (AS_{SING}: 0.744 $[0.711, 0.785]$). Under AS_{fair} the

ID	Claim (short)	Effect AS_{SING} [BCa 95% CI]	AS_{SING}	AS_{fair}
C1/C11	DinoViT best ratio / least steerable (proxy)	$0.149\ [0.142, 0.156]$	holds	opp.
C2	ResNeXt worst ratio (proxy)	$0.323\ [0.303, 0.343]$	holds	opp.
C3/C6m	ResNet > DinoViT (class mean)	$1.383\ [1.372, 1.394]$	holds	opp.
C4	DinoViT least leakage	$-0.293\ [-0.298, -0.289]$	**rev.**	opp.*
C5	ResNeXt top-2 of 5	$1.442\ [1.424, 1.462]$	holds	opp.
C6 sd	ResNet class SD > DinoViT	$0.744\ [0.711, 0.785]$	holds	opp.
C7	DinoViT per-class $\lvert AS\rvert<1°$	$0.500\ [0.499, 0.500]$	holds	n/a
C8 porcupine	Porcupine $\geq$90th pct (ResNet)	$-5.2\ [-8.4, -1.9]$	**rev.**	opp.
C8 sports car	Sports car $\geq$90th pct (ResNet)	$-29.0\ [-34.1, -23.9]$	**rev.**	opp.
C9 spearman	No ResNet–DinoViT class corr.	$-0.033\ [-0.074, -0.012]$	contra.	contra.
C10	Jellyfish $\gg$ camel	$-0.107\ [-0.210, -0.006]$	**rev.**†	opp.
C12	Supp. Fig. 12b order (τ)	$0.821\ [0.56, 1.00]$‡	holds	n.c.‡
C13	EVA02 target-space order	–	untestable	untestable

Table 5: SING's class-level claims under AS_{SING} (BCa 95% CI, images resampled within class, so the CIs are conditional on the 13 fitted translators) and family-A AS_{fair}. rev. = CI on the opposite side of 0; n.c. = not confirmed; opp. = opposite side under AS_{fair}, a diagnostic since AS_{fair} over-corrects; contra. = a nonzero correlation where SING reports none. C1/C11 and C3/C6 mean are one computation each. C1/C2/C11 use $|AS|$ proxies for SING's ratio and steerability. *Under AS_{fair}, C4 coincides numerically with C1/C11 (same model attains the minimum) and is counted once. C6 SD: classes resampled. †Borderline: the percentile interval covers 0. ‡Model bootstrap; under AS_{fair}, $\tau=-0.18\ [-0.78, 0.35]$, not confirmed. C7 has no AS_{fair} analogue.

effect is $-0.203\ [-0.243, -0.166]$, but correcting the SDs for within-class sampling noise shrinks it to -0.025 (AS_{SING}: 0.752), so the AS_{fair} diagnostic for C6 is fragile.

7 Discussion

What we can and cannot conclude. SING's cross-model ranking is indistinguishable from the ranking by null energy. AS responds to null energy through a path that carries no image-specific content, and we can describe that path geometrically. The data cannot separate null energy from real null content as the source of SING's order: content-free surrogates and the content excess both reproduce it, AS per unit energy tracks the order with one translator family and not the other, the class-mean permutation leans toward energy without settling it (its raw numbers are all our pre-registration supports), and no estimator we tried meets all three calibration criteria. We do not conclude that SING's ranking is wrong, or that AS carries no semantic signal; AS responds to planted concepts at fixed energy. The claim we support is narrower: a cross-model AS comparison cannot be read as a comparison of null semantics, because energy and content are not separated by any control we have.

Why does ρ vary? Neural collapse (Papyan et al., 2020) offers a candidate mechanism. Late in training, features concentrate near class means aligned with the classifier rows, so the energy outside the row space shrinks. Models trained further into that regime, or with heads fit differently, would have smaller ρ and hence smaller AS. We have not tested this.

Relation to other work. Similar negative controls appear in other areas. Saliency maps that survive model and data randomisation fail as explanations (Adebayo et al., 2018), attention weights need adversarial and random baselines before they can be read as explanation (Wiegreffe & Pinter, 2019), and apparent emergent abilities can be an artefact of the metric rather than the model (Schaeffer et al., 2023). Our surrogates are the null-space analogue of the surrogate-data test (Theiler et al., 1992): replace the content while keeping the second moments, and see whether the statistic changes. Cui et al. (2022) regress a confound out of representation-similarity scores such as RSA and CKA. Gröger et al. (2026) show that representational similarity metrics are confounded by model scale, calibrate them with permutation nulls, and find that global

convergence largely disappears while local neighbourhood similarity survives, a mixed picture of the same shape as ours. Both argue, as we do, that a cross-model score needs a baseline for a nuisance variable before it is read as a property of the models. Makelov et al. (2024) show that activation patching along a subspace can change behaviour through a dormant pathway that the model does not normally use; their illusion and our energy path both arise when a component the model ignores still moves a readout. In OOD detection, Mahalanobis++ (Müller & Hein, 2025) finds that feature norms confound Mahalanobis scores across 44 models and that ℓ_2 normalisation mitigates it, a feature-energy confound in a different kind of score.

Relation to other null-space work. ActSub (Zöngür et al., 2025) finds that the activation components a classifier largely ignores help to detect distribution shift, and Jacobsen et al. (2019) show that class information can hide in directions the logits do not use. Both are consistent with null content being informative. Cross-model scores of that content, however, need an energy-only baseline. Linear erasure methods (Ravfogel et al., 2020; Belrose et al., 2023) remove a concept so that no linear probe can recover it, a stricter notion than projecting out $\ker W$. A leakage score built on them would ask whether class information is linearly recoverable from the null component rather than how far the translation moves. Linear maps between representations go back to model stitching (Lenc & Vedaldi, 2015; Bansal et al., 2021). Linear translators into CLIP (Moayeri et al., 2023) differ in fidelity across models, and §4.4 shows that the translator affects interventional conclusions.

Recommendations.

1. Report d_{null}, ρ and the null tolerance next to any null-space score, and check rank stability across tolerances.

2. Compare every cross-model AS-type score (a score that measures how far a readout moves when the null component is removed) with the ranking by ρ alone, as a control in the sense of Hewitt & Liang (2019), and with content-free surrogates of matched energy. Report the surrogate used, since the result depends on it.

3. Check conclusions with a second, independently fit translator, and calibrate any content estimator on worlds with known content before using it across models.

4. Name the exact checkpoint and its measured accuracy, and check translator fidelity against a shuffled floor.

5. Flag translator conditioning when full-null and partial-null AS disagree (BiT).

6. Validate flatness or invariance criteria on a synthetic oracle before using them to accept or reject scores (Appendix D).

Limitations. There are only 13 models, so every cross-model CI is wide. SING's order is digitised from a figure. Our retrained translators are ridge maps chosen by us. SING's training recipe may differ, and the equal-energy result depends on this choice. The energy-scan part of the leak-free test covers three models with d_{null}=1048, and the planted strengths stop at α=0.3, below the energy-equivalent strength at each model's own ρ (at ρ_t=0.01, post hoc, content moves AS more than energy does on 2 of the three models). The calibration worlds are semi-synthetic, built on three bases by rescaling class-mean and residual null content; they test whether an estimator can separate the two, not whether real models differ in content. The geometric account is consistent with two held-out models and with synthetic data that builds in its gain law; on the 13 models it is exploratory. The true DINO comparison uses our ridge translators, since SING released none for these models. The energy law does not cleanly separate energy from dimension, and its pre-registered flatness part fails. The class-claim CIs reflect image resampling only (class resampling for C6), several claims rely on proxies, and C13 is untestable. We did not evaluate information-theoretic leakage measures for concept models (Makonnen et al., 2025), which target a different notion of leakage.

8 Conclusion

Using SING's code and released translators, we reproduce its 13-model ranking, and the null energy fraction ρ reproduces it equally well. A content-free path connects null energy to AS, and image-independent null content with each model's own energy reproduces the order. The content excess reproduces it too, and AS per unit energy tracks the order with retrained translators but not with the released ones, so the data cannot separate null energy from real null content as the source of the order. SING's "DinoViT" translator matches a supervised timm checkpoint, and true DINO backbones fall mid-table by raw AS under our ridge translators, with mixed evidence once energy is accounted for. Size-corrected scores change the order without reversing it with confidence. Cross-model AS-type null-space comparisons should report ρ, an energy-only baseline and content-free surrogates of matched energy next to the score.

Reproducibility. The supplementary material contains the code, pre-registrations, aggregated results and a one-command `make reproduce` that regenerates every table, figure and number in this paper from committed aggregates. For review, the code is provided as an anonymised archive; a public link will be added on acceptance. ImageNet-derived features are not redistributed; checksums, the image list and the extraction scripts are included. A full rerun from features takes about 2 h on 8 CPU cores; feature extraction needs a GPU and ImageNet (Appendix E).

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A Pre-registration ledger

Table 6 lists every pre-registered experiment and its outcome. E1 to E9 come from the original plan, R2 from the round-2 pre-registration and R3 from the round-3 one, each written after the previous round's results. Round 4 (R4, R4B and R4C, three files) is in Table 7; the addenda R4B and R4C were written after the R4 results, and each lists the values known when it was written.

Exp.	Pre-declared criterion (short)	Outcome
Exp 0	null size correlates with SING IS/AS	GO rests on ρ (Spearman +0.94 with the leakage order); d_{null} alone +0.46, p=0.12
Exp 1	r=AS(1048)/AS(24) ≥ 3 on $\geq 3/4$; $\tau_{d=24} \leq 0.5$	GO ($r \approx 10$ on 3/4; BiT 0.06; τ=0.29)
E1	high/low $\geq 3\times$ at fixed d; $\leq 1.5\times$ across d at fixed ρ	MIXED (first part 4/4; second fails)
E2	$\tau(\text{AS}_{\text{fair}}, \text{SING}) \leq 0.5$ or headline claim reverses	PASS as registered; weakened in round 2
E3	$\tau(\text{AS}_{\text{fair}}, \text{planted}) \geq 0.8$, $\tau(\text{AS}_{\text{SING}}, \text{planted}) \leq 0.5$	not passed (0.109 vs 0.386); KILL not met
E8	NCP probe $\tau \geq 0.8$ and flat; KILL if $\tau < 0.6$	KILL (τ=−0.331; flat 0/4)
E9	20-candidate screen, DEV $\tau \geq 0.8$ and flat on $\geq 3/4$	KILL (best M1 τ=0.976, flat 0/4)
R2	KILL-R: drop "disagrees" unless fix 2 and D-weak pass	not fired; "differs from" allowed; D-weak screened at the threshold (5/8)
R2	D-strong: τ CI < 0 under $\geq 5/8$ normalisers	FAIL (0/8 all 13; 1/8 screened)
R2	AS_{fair} removes the ρ trend	FAIL on screened set (-0.79 [$-1.00, -0.23$])
R2	tolerance stability: adjacent $\tau \geq 0.9$	FAIL (AS_{fair} 0.87/0.82; AS_{SING} 0.87); $\rho \geq 0.92$
R2	PC bands $G_{\text{top}}/G_{\text{rand}}$ vs 1 at d=24	"top PCs carry less AS per unit energy" (2/3 screened models)
R2	synthetic oracle passes the flatness rule	FAIL (ratio 3.55): rule ill-posed
R2	DinoViT translator fidelity: used checkpoint best by ≥ 0.05	PASS (0.848 vs 0.677); see D
R3-A	equal-energy A1: ENERGY-ACCOUNTS / PARTIAL / KILL-R3	ENERGY-ACCOUNTS ($\tau = -0.03$ E12, 0.09 S11); no KILL
R3-A	A4 (matched CLIP energy) $\tau < 0.3$ on both sets	yes (-0.15, -0.20); $\kappa - \rho$ CI covers 0
R3-B	N4: AS rises with leak-free null energy, all 3 models	PASS (Spearman 1; slopes 0.38–0.69)
R3-B	B-rank: $\tau(\text{AS}_{\text{surr}}) \geq 0.6$ on all 13 and S11	yes (0.69, 0.64); no claim from X
R3-C	PC bands vs energy law $\rho^{\hat{a}}$	"top PCs carry less AS than energy predicts"; not robust in $\hat{a}$
R3-D	N3: non-matching candidates at chance	PASS ($\lvert\cos - \text{shuffled}\rvert \leq 0.002$)
R3-E	N7: no claim CI excludes its point	FAIL (1 flag: AS_{fair} C6 SD); N6 dedup done
R3-F	N1: paired $\tau(\rho) - \tau(\text{AS}_{\text{SING}})$ CI covers 0	yes $[-0.36, 0.29]$: "indistinguishable"
R3-G	retrained translators: order survives if $\tau \geq 0.6$ on all 13 and S11	survives (0.85, 0.82); A1 with retrained translators 0.61, 0.75 (descriptive)

Table 6: All pre-registered experiments of rounds 1–3 and their outcomes. The pre-registration files are in the repository.

Exp.	Pre-declared criterion (short)	Outcome
R4-S	content-free AS_{perm} and content excess $AS_{SING} - AS_{perm}$ each track SING ($\tau \geq 0.6$ on all 13 and S11, CI > 0); KILL-R4-S if $\tau(AS_{perm}) < 0.3$	both track: cannot apportion (0.69/0.64 and 0.62/0.71); surrogate τ spans 0.49–0.72; no kill
R4-J	KILL-R3 rule applied to retrained A1 (reading only); influence of DinoViT/EVA02	retrained: KILL-R3 level (0.61, 0.75); released S11 without the pair 0.61
R4-J	A4 under ridge translators < 0.3 on both sets	no (0.52, 0.49): holds for released translators only
R4-DN	KILL-DN (feature validity); Q-DN1 (DINO among lowest 4 AS); Q-DN2 (inside energy-law PI)	no kill; Q-DN1 SING-inconsistent (ranks 8, 9 of 15); Q-DN2 mixed
R4-P	energy-equivalent planted strength α^*	$\alpha^* > 0.3$ on all 3 models (outside the tested range)
R4-C6	C6-SD point inside both CIs (class resampling); no flags	PASS
R4-N10	fast holdout < 30 min and within ± 0.03	FAIL (36.3 min; ΔAS_{fair} nl 0.057); not re-timed, see Appendix E
R4B-1	class-mean permutation: CONTENT-CARRIES / ENERGY-CARRIES / UNAPPORTIONED; KILL-R4B-1 on CONTENT-CARRIES	ENERGY-CARRIES under both families (released 0.64/0.64; ridge 0.70/0.67); no kill
R4B-2	semi-synthetic worlds C/M/D: content-diagnostic estimator	none of 8 estimators diagnostic on all 3 bases
R4B-5	NCM_{null}: NCM-CONTENT (≥ 0.6, partial CI $\not\ni 0$) / NCM-NULL (< 0.3)	intermediate (0.45, 0.42); no kill
R4B-3	selectivity R: R-TRACKS (≥ 0.6) / R-NULL (< 0.3); within-model CI $\not\ni 1$ on $\geq 9/12$	R-NULL (0.15, 0.05); within-model 12/12 (4 above, 8 below 1)
R4B-6	DINO inside the E12 PI of Δ_{align} and NCM_{null}; both in the lowest 4	inside both PIs for both; 'least leaking' label not met (it needs both models in the lowest 4); ViT-S/16 alone is in the lowest 4 on both (ranks 3 of 15 and 2 of 14)
R4C-G2	geometry on true DINO: (i) Spearman ≥ 0.9, (ii) oop CI > 0, (iii) closed form $\leq 25\%$; (iv) $\lvert a_{fit} - a_{pred} \rvert \leq 0.1$	PASS on both; exponent predicted; no kill
R4C-G3	synthetic: a_{fit} within ± 0.05 of $(1+s)/2$	PASS (max error 0.027)
R4C-B3	NZS gate (rescaling $\leq 5\%$, planted $\tau \geq 0.6$); energy-free $\lvert \tau(\cdot, \rho) \rvert < 0.3$; SING test	gate pass; not energy-free (0.45, 0.53); SING test inconclusive
R4C-B4	IS energy proxy: $\lvert \tau(\mathrm{IS}, 1-\rho) \rvert \geq 0.6$ on all three sets	no (-0.62, -0.55, -0.49)

Table 7: Round-4 pre-registered tests and outcomes (E12 / S11 unless stated). R4B and R4C rows come from addenda written after the R4 results. No kill fired.

B Additional tables

Model	d_{null}	ρ	κ	γ	AS_{SING}	A1	AS_{surr}	fair A	fair B	B max Δlogit/SD
VGG16	3096	0.350	0.119	0.34	1.088	0.084	0.922	0.708	0.867	14.4
VGG19	3096	0.339	0.123	0.36	1.256	0.094	0.879	0.716	1.087	13.7
DenseNet121*	27	1.8×10^{-8}	2.5×10^{-8}	1.36	2.5×10^{-4}	–	2.4×10^{-4}	0.152	0.671	1.5×10^{-8}
ResNet50	1048	0.146	0.281	1.93	1.676	0.160	1.176	0.855	0.989	0.3
"DinoViT"	24	0.005	0.013	2.46	0.294	0.284	0.296	1.632	1.220	5.0×10^{-3}
EffNet-B0	280	0.130	0.030	0.23	0.443	0.076	0.378	0.889	1.267	0.8
BiT-R50x1*	1048	0.046	0.061	1.31	0.557	0.153	0.589	0.042	1.142	9.3×10^{-2}
ResNeXt101	1048	0.451	0.219	0.49	1.999	0.112	1.476	0.707	0.591	74.4
ConvNeXt-B	24	0.013	0.002	0.17	0.080	0.049	0.085	0.974	0.961	4.1×10^{-2}
Swin-L	536	0.274	0.151	0.55	1.019	0.105	1.170	0.829	1.130	7.5
DINOv2-L	1048	0.322	0.165	0.51	1.499	0.104	0.960	0.712	0.961	6.0
DeiT3-L	24	0.348	1.085	3.11	5.048	0.215	3.364	1.065	0.776	16.4
EVA02-L	24	0.009	0.031	3.47	0.417	0.302	0.411	2.612	1.306	2.3×10^{-2}

Table 8: Per-model scores on the 10k subset at tolerance 10^{-5}. κ: CLIP-space null energy; $\gamma = \kappa/\rho$; A1: AS after removing $\sqrt{\rho_0/\rho}\,\Pi f$; AS_{surr}: AS with leak-free surrogate null content; fair A/B: AS_{fair} families; last column: largest logit change of the family-B control in logit SDs (family B is not logit-preserving). *Excluded by the screen. Family C: Table 11.

Model	AS at α=0, ρ_t = 0.01	0.05	0.2	0.45	slope [95% CI]	AS(α)/AS(0), ρ_t=0.2 α=0.1	α=0.3
ResNet50	0.371	0.665	0.817	1.935	0.377 [0.375, 0.379]	1.088	1.508
ResNeXt101	0.074	0.175	0.453	1.105	0.690 [0.689, 0.692]	1.040	1.353
DINOv2-L	0.082	0.192	0.485	1.151	0.677 [0.676, 0.678]	1.049	1.363

Table 9: Leak-free null content (§4.2). AS at α=0 against the surrogate's null energy ρ_t; OLS slope of logAS on $\log \rho_t$ with 1000× image bootstrap; AS response to a planted concept at ρ_t=0.2.

Model	d=24, $\hat{a}$ = 0.431	0.388	0.485	d=100, $\hat{a}$ = 0.431	0.388	0.485
ResNet50	1.50 [1.48, 1.51]	1.75 [1.73, 1.77]	1.23 [1.21, 1.24]	1.18 [1.17, 1.19]	1.30 [1.28, 1.31]	1.05 [1.03, 1.06]
ResNeXt101	0.74 [0.73, 0.76]	0.85 [0.83, 0.87]	0.63 [0.61, 0.64]	0.97 [0.96, 0.99]	1.06 [1.04, 1.07]	0.87 [0.86, 0.89]
DINOv2-L	0.98 [0.97, 0.99]	1.06 [1.04, 1.07]	0.89 [0.88, 0.90]	0.91 [0.89, 0.92]	0.96 [0.95, 0.97]	0.85 [0.83, 0.86]
BiT-R50x1*	0.009 [0.009, 0.009]	0.010 [0.010, 0.010]	0.008 [0.008, 0.008]	0.012 [0.012, 0.012]	0.013 [0.013, 0.013]	0.011 [0.011, 0.011]

Table 10: PC bands against the fitted energy law (§4.5): H with 1000× image-bootstrap CIs at three values of $\hat{a}$ (E1 point estimate and CI ends). $H < 1$: top PCs carry less AS than energy predicts. *BiT is reported but not counted.

	VGG16	VGG19	Dense	RN50	DinoViT	Eff	BiT	RNX	ConvNeXt	Swin	DINOv2	DeiT3	EVA02
fair C	1.32	1.35	0.71	1.14	1.20	1.44	135.6	1.04	0.97	1.08	1.00	0.79	1.33

Table 11: Family-C $\mathrm{AS}_{\mathrm{fair}}$ per model (`r2/results/fix2_models.csv`).

Kind	Null content	all 13	E12	S11	partial r given ρ (S11)
content-free	$\mathcal{N}(\mu, \Sigma)$ (round 3, $\mathrm{AS}_{\mathrm{surr}}$)	0.69 [0.39, 0.92]	0.64 [0.31, 0.90]	0.64 [0.27, 0.92]	0.52 [−0.07, 0.94]
content-free	$\mathcal{N}(0, \Sigma)$	0.72 [0.43, 0.94]	0.67 [0.38, 0.93]	0.67 [0.35, 0.96]	–
content-free	isotropic $\mathcal{N}(0, I)$	0.49 [0.04, 0.89]	0.39 [−0.04, 0.87]	0.60 [0.16, 0.96]	–
content-free	mean only (μ)	0.49 [0.07, 0.83]	0.39 [−0.05, 0.74]	0.38 [−0.16, 0.79]	–
content-free	real c permuted across images, $\mathrm{AS}_{\mathrm{perm}}$	0.69 [0.39, 0.92]	0.64 [0.31, 0.90]	0.64 [0.27, 0.92]	0.52 [−0.07, 0.94]
class content	real c permuted within class	0.79 [0.50, 1.00]	0.76 [0.46, 0.97]	0.78 [0.44, 1.00]	–
excess	$X = \mathrm{AS}_{\mathrm{SING}}/\mathrm{AS}_{\mathrm{surr}}$	0.54 [0.14, 0.83]	0.58 [0.21, 0.86]	0.60 [0.24, 0.91]	0.79 [0.02, 0.97]
excess	$\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{surr}}$	0.59 [0.15, 0.89]	0.64 [0.24, 0.90]	0.67 [0.27, 0.96]	0.81 [0.02, 0.98]
excess	$\mathrm{AS}_{\mathrm{SING}} - \mathrm{AS}_{\mathrm{perm}}$	0.62 [0.15, 0.92]	0.67 [0.25, 0.96]	0.71 [0.28, 1.00]	0.77 [0.04, 1.00]
excess	within-class/across-image	0.41 [0.00, 0.69]	0.45 [0.12, 0.74]	0.45 [0.09, 0.75]	0.57 [−0.48, 0.93]
–	real ($\mathrm{AS}_{\mathrm{SING}}$)	0.82 [0.56, 1.00]	0.79 [0.51, 1.00]	0.82 [0.51, 1.00]	0.78

Table 12: Surrogate null content (§4.2): Kendall τ against SING [percentile 95% CI], released translators, 3 draws per surrogate (5 for the Gaussian), each rescaled to the model's own null energy. Last column: partial Spearman given ρ on S11 with its CI.

Model	own ρ	AS(α)/AS(0), ρ_t=0.2 α=0.1	α=0.3	AS(α)/AS(0), own ρ α=0.1	α=0.3	AS (°) own ρ, α=0/0.3	r_E
ResNet50	0.146	1.088	1.508	1.099	1.581	0.768/1.214	2.39
ResNeXt101	0.451	1.040	1.353	1.018	1.137	1.107/1.259	5.37
DINOv2-L	0.322	1.049	1.363	1.019	1.169	0.762/0.891	4.92

Table 13: Planted content (§4.2): AS response to a planted concept at the surrogate energy ρ_t=0.2 and at each model's own ρ, absolute AS at its own ρ, and $r_E = \mathrm{AS}(\rho_{75})/\mathrm{AS}(\rho_{25})$, the AS change the energy law gives across the interquartile range of ρ.

Estimator	World C: $\lvert\tau(\cdot,\rho)\rvert \le 0.2$			World M: $\tau(\cdot,L) \ge 0.8$			World D: $\tau(\cdot,L) \ge 0.6$			
	RN50	RNX	DINOv2	RN50	RNX	DINOv2	RN50	RNX	DINOv2	diagnostic
ρ	1.00	1.00	1.00	1.00	1.00	1.00	–	–	–	no
AS	1.00	1.00	1.00	0.33	1.00	1.00	0.24	0.97	0.94	no
A1	0.70	0.55	0.00	0.45	−0.76	−0.85	−1.00	0.48	0.94	no
$\mathrm{AS}_{\mathrm{gauss}}$	1.00	1.00	1.00	0.94	1.00	1.00	−0.55	−1.00	−0.94	no
X	−0.52	−1.00	−1.00	−0.67	−0.76	−0.88	0.97	1.00	0.97	no
Δ_{align}	−1.00	−1.00	−1.00	−0.15	0.61	0.18	0.94	1.00	0.94	no
$\mathrm{NCM}_{\mathrm{null}}$	−0.32	−1.00	−1.00	0.92	0.88	0.80	0.93	1.00	1.00	no
selectivity R	−0.88	−0.45	−0.30	−0.06	−0.85	0.79	−0.73	−0.88	−0.48	no
exponent a	0.55	0.49	0.30	0.28	0.52	0.42		–		

Table 14: Calibration worlds (§4.6): Kendall τ against ρ in world C and against L (class-mean null energy as a share of total feature energy) in worlds M and D, on three bases. Criteria in the header; "diagnostic" requires all nine. Last row: fitted energy exponent.

Model	AS	$\mathrm{AS}_{\mathrm{in}}$	$\mathrm{AS}_{\mathrm{oop}}$	oop share	oop closed-form err.	$a_{\mathrm{pred}} = (1+s)/2$	a_{fit}
Exploratory (post hoc): 13 models, released translators							
VGG16	1.088	0.786	0.667	0.61	0.086	0.42	0.36
VGG19	1.256	0.857	0.678	0.54	0.086	0.42	0.36
ResNet50	1.676	0.938	1.645	0.98	0.160	0.40	0.44
"DinoViT"	0.294	0.291	0.096	0.33	0.016	0.33	–
EffNet-B0	0.443	0.395	0.201	0.45	0.024	0.69	0.67
BiT-R50x1	0.557	0.502	0.405	0.73	0.052	0.08	0.40
ResNeXt101	1.999	1.169	1.287	0.64	0.116	0.44	0.45
ConvNeXt-B	0.080	0.081	0.016	0.20	0.002	0.79	–
Swin-L	1.019	0.938	0.849	0.83	0.096	0.41	0.45
DINOv2-L	1.499	0.857	1.096	0.73	0.124	0.29	0.39
DeiT3-L	5.048	1.926	4.626	0.92	0.185	0.37	–
EVA02-L	0.417	0.393	0.229	0.55	0.044	−0.03	–
Confirmatory (pre-registered G2): true DINO, ridge translators							
DINO ViT-B/16	0.514	0.384	0.267	0.52	0.033	0.60	0.67
DINO ViT-S/16	0.652	0.630	0.519	0.80	0.066	0.57	0.58
Synthetic calibration (G3): known gain slope s							
s	−0.4, −0.2, 0.0, 0.2				$a_{\mathrm{fit}} - (1+s)/2$:	−0.002, −0.004, 0.006, 0.027	

Table 15: Geometric decomposition of AS (§4.7), degrees. oop: out-of-plane; closed-form error: relative error of $\frac{1}{2}\cot\theta\|\delta_{\mathrm{oop}}\|^2/\|T(f)\|^2$ among images with $\|\delta_{\mathrm{oop}}\|/\|T(f)\| < 0.5$; a_{fit}: energy exponent fitted on matched-d cells (– where the cell fit is not available). DenseNet omitted ($\rho \approx 0$).

C Energy law, Exp 1 and checkpoint accuracy

Model	r	2σ band	R^2 (vs $d \equiv$ vs ρ_d)
ResNet50	10.15	[5.83, 14.48]	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512

Table 16: Exp 1: $r = \mathrm{AS}(d{=}1048)/\mathrm{AS}(d{=}24)$ over random null subspaces (20 seeds). For random subspaces ρ_d is linear in d, so the two R^2 coincide. At matched $d{=}24$ across 8 models, τ against SING is 0.29 ($p{=}0.40$).

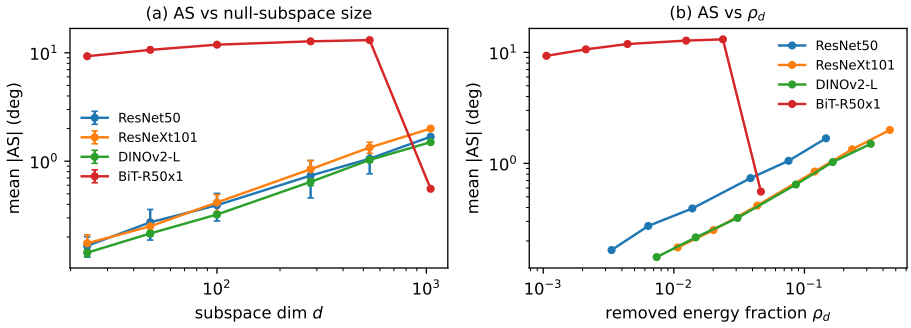


Figure 3: Exp 1: mean |AS| against subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation). Full-null anchors: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°.

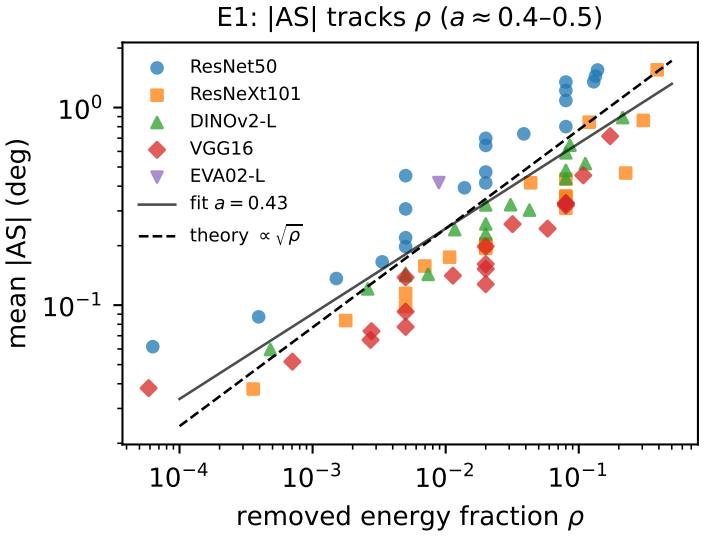


Figure 4: Energy law: |AS| against ρ over 76 cells on five models (ResNet50, ResNeXt101, DINOv2-L, VGG16, EVA02-L) at fixed $d \in \{24, 100, 280\}$ and fixed $\rho \in \{0.5\%, 2\%, 8\%\}$. Solid: pooled fit ($a$=0.431); dashed: $\sqrt{\rho}$. At fixed d the high/low-energy ratio exceeds 3× on all 4 evaluable models; at fixed ρ the max/min across d exceeds 1.5 in several cells (up to 2.29), so the pre-registered rating is *mixed*.

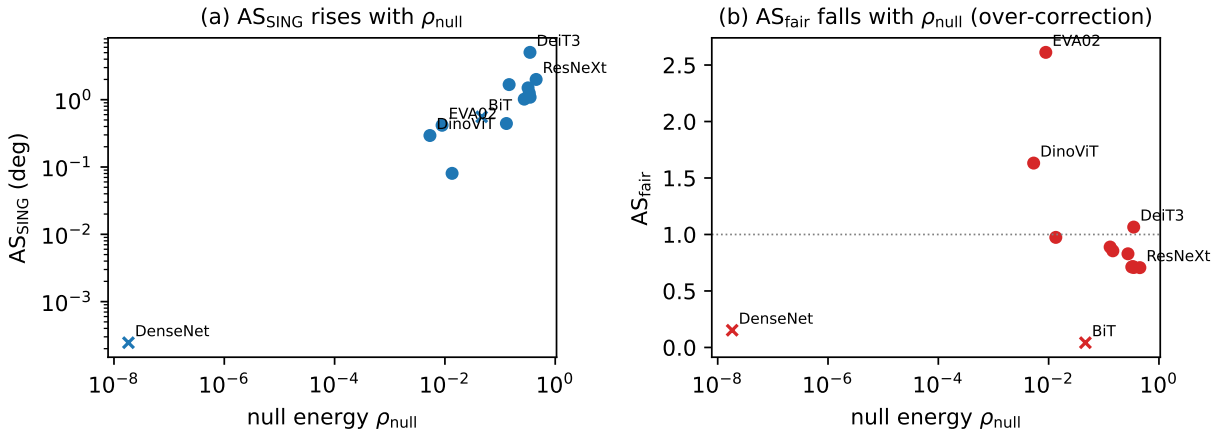


Figure 5: (a) AS_{SING} rises with ρ. (b) Family-A AS_{fair} falls with ρ: it over-corrects. $\times$: models excluded by the screen.

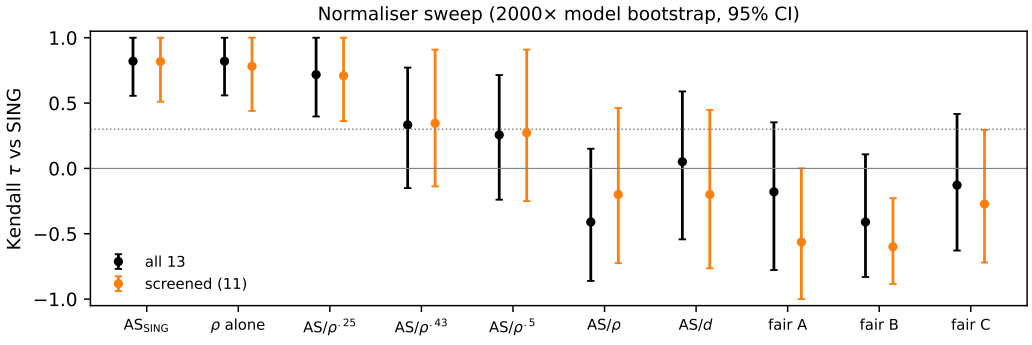


Figure 6: τ against SING for each normaliser and for ρ alone (2000× model bootstrap), on all 13 models and on S11. Dotted line: the τ=0.3 bar.

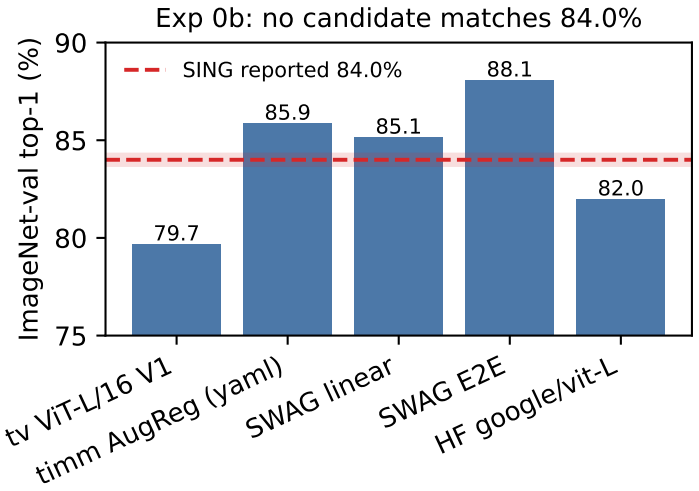


Figure 7: Top-1 on ImageNet-val (native evaluation transform) of the five public ViT-L/16 candidates. None matches SING's reported 84.0%.

D Planted leakage, score search and the flatness rule

Planted leakage (E3). We planted a CLIP concept direction (`red`, `striped`, `water`) into the null spaces of 8 models at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, with logits unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ against the planted ranking, averaged over concepts, is 0.386 $[0.324, 0.448]$ for AS_{SING} and 0.109 $[0.046, 0.171]$ for AS_{fair} (both α=0-subtracted). The pre-registered target ($\text{AS}_{\text{fair}} \geq 0.8$, $\text{AS}_{\text{SING}} \leq 0.5$) was not met. Neither score recovers planted strength well across models.

E8 (pre-registered; killed). A probe-based null-content score (ridge-probe R^2 on null coordinates, z-scored against energy-matched random subspaces) gave $\tau{=}-0.331$ $[-0.392, -0.273]$ on the planted task and was flat on 0/4 models. Three implementation caveats apply: the z-score saturates, λ was chosen by approximate leave-one-out rather than the pre-registered exact cross-validation, and λ sat at the grid floor in 18 of 96 fits.

E9 (pre-registered; killed). We screened 20 candidate scores on two development concepts. The rule required $\tau \geq 0.8$ and flatness (max / min ≤ 1.5 across nested null subspaces) on at least 3 of 4 models at $\alpha \in \{0.1, 1\}$. No candidate passed. The best was M1, a debiased null-leakage estimate $\hat{L}^2 = (\|\sum_i Pf_i s_i\|^2 - \sum_i \|Pf_i\|^2 s_i^2)/(n(n{-}1)\mathbb{E}\|f\|^2)$ with standardised concept scores s_i, scored as $\sqrt{\max(\hat{L}^2, 0)}$. It reached $\tau{=}0.976$ $[0.963, 0.988]$ but was flat on 0/4 models.

The flatness rule is ill-posed. We tested the rule on synthetic Gaussian features with a known head ($m{=}400$, $d_{\text{null}}{=}300$). Our exact oracle is the planted amplitude inside the removed subspace. It fails the rule with max / min 3.55 and 3.48 ($\approx \sqrt{300/24}$), because a larger removed subspace contains more of the plant. A revised rule (max / min of score/oracle ≤ 1.5, post hoc) is passed by M1 (ratios 1.00–1.03). Under the revised rule M1 is flat on 4/4 models, and E8's score on 0/4. E8 and E9 remain killed as registered.

Holdout (post hoc). We ran the E9 holdout on four held-out concepts (water, green, thin, knitted; 50 trials at seed 1; linear and nonlinear plants) for the six candidates declared in the round-2 pre-registration plus AS_{SING} and AS_{fair}, with the E9 pass rule (Table 17). Only M1 meets both τ thresholds. Under the registered flatness rule no candidate passes; that rule also fails for the oracle (above). Under the revised rule, M1 passes. Paired over trials (1000×, seed 0), $\tau(\text{M1}){-}\tau(\text{AS}_{\text{fair}})$ is 0.77 [0.71, 0.82] for the linear plant and 1.01 [0.96, 1.06] for the nonlinear plant, so M1 tracks the planted leakage better than AS_{fair} under both

Score	lin. τ [95% CI]	nl. τ	flat	rev. flat	pass
M1	0.93 [0.91, 0.95]	0.97	0/0	4/4	rev.
M2	0.44 [0.39, 0.48]	0.46	0/0	0/0	no
S9	0.45 [0.40, 0.51]	0.44	0/1	2/0	no
S10	0.37 [0.33, 0.42]	0.52	3/4	0/0	no
M5 ($\mathrm{AS}/\sqrt{\rho}$)	0.22 [0.18, 0.27]	0.15	0/0	0/0	no
S4 (AS/ρ)	−0.08 [−0.15, −0.01]	−0.14	0/0	0/0	no
$\mathrm{AS}_{\mathrm{SING}}$ (raw)	0.33 [0.27, 0.40]	0.36	0/0	0/0	no
$\mathrm{AS}_{\mathrm{fair}}$ (sub.)	0.17 [0.11, 0.22]	−0.04	0/0	0/0	no

Table 17: E9 holdout (post hoc; $\mathrm{AS}_{\mathrm{fair}}$ is α=0-subtracted; water, green, thin, knitted; 50 trials, seed 1; 8 models). Flat = number of the 4 large-null models flat at α=0.1/1 under the registered rule; rev. flat = under the revised score/oracle rule. Pass: lin. $\tau \geq 0.8$, CI lower ≥ 0.7, nl. $\tau \geq 0.6$, flat on $\geq 3/4$ at both α (yes = registered rule, rev. = only with the revised rule).

plants. The revised rule also changes which candidate counts as flat: S10, the only candidate flat under the registered rule (3/4 and 4/4 models at α=0.1/1), is flat on 0/4 and 0/4 under the revised one. The plant is written along the null projection that M1 itself measures, and M1, unlike the AS-family scores, is given the concept score that defines the plant. This comparison therefore favours M1 by construction. One of the four held-out concepts, water, was also an E3 plant concept whose results we had seen before choosing the holdout set; its inclusion was pre-registered, but the set is not fully unseen. M1 was picked after the DEV results and the flatness rule was revised after the oracle test, so this holdout is post hoc. E9 stays a KILL as registered, and M1 is a candidate for a future pre-registered test, not a validated score.A reduced-draw mode (5 instead of 20 random draws for $\mathrm{AS}_{\mathrm{fair}}$, 2 instead of 5 nested bases, S9 omitted), declared before it was run, gives the same M1 and $\mathrm{AS}_{\mathrm{SING}}$ values and the same flatness verdicts. $\mathrm{AS}_{\mathrm{fair}}$ moves by +0.02 (linear) and +0.06 (nonlinear; 0.057), and the run takes 36 min on 8 CPU cores. It fails both targets we set ($\pm$0.03, under 30 min), so the full run (1.95 h) is the reference (Appendix E).

M1 on the 13 models (post hoc, descriptive). At α=0 on the full null, averaged over SING's 173 concepts, M1 gives τ=0.667 $[0.29, 0.94]$ against SING, an interval that includes $\mathrm{AS}_{\mathrm{SING}}$'s value. M1 orders the models much as ρ does (τ=0.79), as expected for a score scaled by total feature energy. Its holdout validation favours it by construction (the plant is written along the projection it measures), so this agreement is not independent evidence about energy. We treat M1 as a candidate for a future pre-registered test, not as a validated score.

E Reproducibility and compute

`make reproduce` regenerates every table, figure and number in this paper from committed aggregates in under a minute. This checks consistency, not the feature-level computation. `make reproduce-full` reruns the feature-level analyses from the private ImageNet features: about 2 h wall on 8 CPU cores, most of it the holdout. `make holdout-fast` runs the reduced holdout (36 min on 8 cores). It fails its pre-registered accuracy target on a deterministic quantity, so a faster machine cannot make it pass, and we did not re-time it. The timing itself may be affected by contention: the machine is shared, an earlier attempt was aborted because other jobs were running, the load average was not recorded for the timed run, and the job used about 7.3 of the 8 cores. The round-4 analyses (`r4/`) run on CPU from the same private features (`make reproduce-r4`); extracting the true DINO features needs a GPU. Feature extraction (13 models on the 10k subset, the Exp 1 features and the five ViT-L/16 candidates on the 50k validation set) needs ImageNet and a GPU. Two jobs were timed: the Exp 1 extraction (1.4 GPU-hours on one cloud GPU) and the re-extraction of the five ViT-L/16 candidates for the translator check (1.9 h on two T4 GPUs, about 3.8 GPU-hours including weight downloads). The other extraction jobs were not timed; we estimate under 10 GPU-hours for all extraction.

Seeds. Family A random draws 10000+r ($r < 20$); family B 50000+r; family C subspaces 30000+j and controls 60000+j; Exp 1 20 seeds per d; E3 plants seed 0, bootstrap 10000× seed 1; model bootstrap 2000× seed 0; claim bootstrap 1000× seed 0; E9 and holdout α seed 1; flatness bases 20000+j; PC-band draws

40000+j; A3 rotations 80000+j; leak-free surrogates 70000+r; translator split permutation seed 0. Round 4: surrogates 92000+r (zero-mean, isotropic, permuted) and 93000+r (within-class permutation); class-mean derangements 94000+r; NCM shuffled labels 95000; selectivity random rotations 96000+j; NZS control 97000; synthetic geometry model 98000; geometry random subspaces 1000d+r; selectivity and geometry image bootstraps seed 0.